

Wang Paper Model Notes

Quantifying the Dynamics of Failure Across Science, Startups, and Security

Wang, Yin, & Wang (2019)

1. Central Question of the Paper

The paper asks:

Why do some individuals eventually succeed after repeated failures, while others continue failing indefinitely?

More deeply, the authors ask whether: - failure trajectories contain hidden structural information, - repeated attempts obey universal statistical laws, - and common mechanisms govern eventual success across radically different domains.

This is important because the paper is NOT merely: - describing success, - predicting outcomes, - or measuring persistence.

Instead, the paper attempts to:

identify a universal generative mechanism underlying repeated failure and eventual success.

That ambition is extremely important conceptually.

The paper is fundamentally a: - complex systems paper, - mechanism discovery paper, - and cross-domain theory paper.

This is one of the strongest parallels to the current advancement project.

2. Domains Studied

The authors analyze three very different systems:

Domain	Success Event	Failure Event
Science	successful NIH grant	rejected grant applications
Entrepreneurship	startup acquisition / IPO	failed ventures
Security / terrorism	successful attacks	failed attacks

This cross-domain design is one of the paper's most important contributions.

The authors intentionally choose: - socially, - economically, - organizationally, - and behaviorally

different systems.

The argument is:

if similar dynamics appear across these domains, then the mechanism may be fundamental rather than

This is directly analogous to: - Army promotion, - NBA draft systems, - and academic tenure systems

in the current project.

3. Central Empirical Observation

One of the paper's key empirical findings:

eventual successes and perpetual failures often look surprisingly similar early on.

This is extremely important.

The authors show: - eventual successful actors frequently experience long failure streaks, - and raw failure counts alone do not explain eventual success.

Instead, the important distinction becomes:

how trajectories evolve through repeated attempts.

The paper finds that eventual successful actors: - systematically improve, - iterate more efficiently, - and increasingly refine future attempts.

Perpetual failures, by contrast: - exhibit weaker adaptation dynamics, - slower improvement, - and less cumulative refinement.

This is one of the deepest conceptual moves in the paper:

failure itself is not diagnostic; trajectory structure is.

4. Failure Streak Dynamics

The paper observes that:

failure streaks are heavy-tailed.

Interpretation: - some actors fail only briefly, - others fail repeatedly for extremely long periods.

Importantly: - the process is NOT memoryless, - and failure sequences are NOT purely random.

This matters because: - if failures were random independent events, then: - streak lengths would follow much simpler distributions.

Instead: - the persistence structure itself contains information about the underlying process.

This is one of the first major clues that: - hidden learning/adaptation dynamics exist.

5. The Exploration–Exploitation Tradeoff

One of the paper’s central conceptual mechanisms is:

exploration *vs* exploitation

The authors model each attempt as composed of: - reusable components, - and modifiable/new components.

Actors face a tradeoff: - exploit previously successful elements, OR: - explore new alternatives.

This is a classical complex systems / adaptive systems problem.

The key insight is:

successful trajectories balance reuse and innovation efficiently.

Too much exploration: - creates instability, - prevents cumulative improvement, - and sacrifices retained gains.

Too much exploitation: - traps actors in stagnant local optima, - and prevents meaningful improvement.

The paper argues: - successful actors dynamically manage this balance better over time.

This idea is deeply important because: - success emerges not from static superiority, but from: - adaptive trajectory dynamics.

6. The k-Memory Model

The paper introduces a minimal mechanistic model.

Core idea:

Actors remember only:

k

previous attempts.

This parameter controls: - how much prior information can be reused, - retained, - refined, - or recombined.

Each new attempt partially: - reuses prior successful components, while: - replacing weaker components.

The model is intentionally simple.

This is important.

The authors are NOT trying to: - perfectly simulate reality, - reproduce all institutional details, - or maximize realism.

Instead, the goal is:

identify the smallest mechanism set capable of reproducing the observed macro-level phenomenon.

This modeling philosophy is extremely relevant to the current Tier 1 advancement framework.

7. Critical Threshold / Phase Transition

One of the paper’s most important results:

There exists a critical threshold:

$$k^*$$

such that system behavior changes qualitatively.

Region	Outcome Dynamics
$k < k^*$	stagnation / persistent failure
$k > k^*$	systematic improvement and eventual success

This is effectively a: - phase transition, - tipping point, - or regime shift.

Tiny changes in: - adaptive capacity, - memory, - or refinement dynamics

can produce radically different long-run trajectories.

This is conceptually very important for the advancement project because: - nonlinear advancement effects, - congestion thresholds, - or distinction-allocation tipping points

may eventually exhibit similar behavior.

8. Temporal Scaling Laws

The paper discovers scaling laws governing: - timing, - iteration, - and performance dynamics.

One key result:

$$T_n \sim n^{-\gamma}$$

where: - T_n = time between attempts, - and successful actors compress inter-attempt intervals over time.

Interpretation: - successful actors iterate faster, - learn more efficiently, - and refine trajectories cumulatively.

This creates accelerating adaptation.

Importantly: - the paper links macro-level trajectory patterns to: - simple underlying behavioral rules.

This is a hallmark of strong complex systems modeling.

9. Structural Rather Than Moral Explanation

One of the paper's strongest conceptual contributions:

The authors avoid: - simplistic merit narratives, - purely psychological explanations, - and "successful people are simply better" arguments.

Instead, the paper argues:

structural adaptive dynamics generate divergence over time.

This is important philosophically.

The paper reframes: - success, - persistence, - and failure

as: - emergent trajectory properties.

This is strongly analogous to the advancement project's emerging logic:

advancement outcomes may emerge structurally from constrained distinction systems.

rather than: - purely individual merit alone.

10. Why This Paper Matters for the Advancement Project

The parallels are substantial.

The Wang paper asks:

Can common structural mechanisms govern success trajectories across radically different domains?

The current project asks:

Can common structural mechanisms govern advancement dynamics across radically different domains

Both projects: - identify cross-domain regularities, - search for universal mechanisms, - and use minimalist mechanistic modeling.

11. Strong Conceptual Parallels

Wang Paper	Advancement Project
repeated failure	repeated advancement competition
adaptive learning	developmental benefit
exploration/exploitation	development vs congestion tradeoff
phase transition	inverted-U turning point
universal mechanism	cross-domain advancement theory
minimal mechanism model	Tier 1 framework
structural divergence	structural advancement distortion

These parallels are conceptual rather than literal.

The exact mathematics will likely differ.

However, the: - scientific style, - explanatory philosophy, - and modeling discipline
are highly relevant.

12. Most Important Modeling Lesson

The most important lesson is NOT: - the exact equations, - or the exact failure dynamics machinery.

The most important lesson is:

minimal interpretable models can reveal surprisingly general structural mechanisms.

The Wang paper succeeds because: - the model is simple enough to interpret, yet: - rich enough to reproduce important empirical regularities.

This is likely the correct direction for: - Tier 1 advancement modeling.

13. What Should Probably Be Borrowed

Strong candidates: - minimal mechanism philosophy, - cross-domain replication strategy, - phase-transition thinking, - interpretable parameterization, - simulated counterfactual worlds, - mechanism isolation.

Especially important:

the model is designed to explain the phenomenon, not merely fit the data.

That distinction is critical.

14. What Should NOT Be Borrowed Too Literally

The current advancement framework is fundamentally different from: - repeated attempt refinement.

Your framework is much more about: - constrained distinction allocation, - local competition, - finite opportunity, - evaluative congestion, - and advancement bottlenecks.

Therefore: - the exact Wang machinery should NOT be copied directly.

The value lies more in: - the scientific approach, - the minimalist philosophy, - and the cross-domain reasoning strategy.

15. Most Important Meta-Level Takeaway

The Wang paper demonstrates that:

complex macro-level human outcomes may emerge from surprisingly simple underlying structural rule

This is perhaps the single most important lesson for the advancement project.

The paper's success comes from simultaneously: - identifying empirical regularities, - constructing mechanistic explanations, - and reframing the phenomenon theoretically.

That is very close to the current direction of the advancement research effort.

16. Final Synthesis

The Wang paper is fundamentally a: - mechanism discovery paper, - complex systems paper, - and universality paper.

Its deepest contribution is demonstrating that: - repeated failure trajectories, across: - science, - entrepreneurship, - and security,

may arise from: - shared structural adaptive dynamics.

For the current project, the key takeaway is:

minimal structural models with strong explanatory power are often more scientifically valuable than hi

This is likely the correct guiding philosophy for: - Tier 1 advancement modeling, - early simulation worlds, - and mechanism validation.

Citation

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